

Candidate E032

Synthetic candidate profile

EMPLOYMENT HISTORY

Senior Big Data Engineer | Midwest Manufacturing Solutions | 2018-03 to Present

Currently leading the design and implementation of a cloud-native data lake on AWS, leveraging BigQuery and Airflow for real-time ingestion and transformation. My focus is on ensuring data quality and performance at scale, mentoring junior engineers, and driving the adoption of best practices across the data platform.

Big Data Engineer | Midwest Manufacturing Solutions | 2015-02 to 2018-02

Moved from the operations research team to the data engineering group, where I built ETL pipelines and introduced NiFi for data ingestion. I was responsible for migrating on-premise data warehouses to a Hadoop-based data lake, which improved processing times by 40%.

Operations Research Analyst | Midwest Manufacturing Solutions | 2010-01 to 2015-01

Initially worked on production scheduling and supply chain optimization using statistical modeling. I began automating data collection and reporting with Python and SQL, which eventually led me to the data engineering world and sparked my passion for building scalable data solutions.

SKILLS

ETL pipelines; Google BigQuery; Apache Airflow; Apache NiFi; Data lake architecture; Data validation; Performance tuning; Data quality monitoring; Data ingestion; Apache Cassandra; Apache Flink; Apache HBase; Amazon S3; dbt; Apache Hive; Databricks; Python; SQL

EDUCATION

PhD | Industrial Engineering | Iowa State University

2005-09 to 2010-06

PROJECTS

Real-time Quality Monitoring

Designed and deployed a stream processing pipeline using Flink and Kafka to monitor sensor data from manufacturing lines, reducing defect detection time from minutes to seconds.

Technologies used: Apache Flink, Apache Kafka, Amazon S3.

Data Lake Migration

Led the migration of legacy data warehouses to a data lake on AWS S3, using Hive for batch processing and implementing a medallion architecture with dbt for transformations.

Technologies used: Amazon S3, Apache Hive, Databricks.

CERTIFICATIONS

Google Professional Data Engineer

Issued by Google.

2019-01

Databricks Certified Associate Developer for Apache Spark

Issued by Databricks.

2021-01